COURSE 5 · ASTRONOMY

Astronomy

Theory reader, audio edition

The sky, gravity and orbits, light, the Solar System, the lives and deaths of stars, galaxies and the history of the universe.

8 modules · equations spoken in words; numbered equations appear in the Visual Companion.

Before you start

How to use this reader

This is the audio edition. It is written to be listened to in a text-to-speech app as comfortably as read on the page. Equations are spoken in words, and each important one carries a number like this: [Equation 5.1-1]. The same number appears in the Visual Companion, where the equation is printed in ordinary notation.

Each module takes about twenty to thirty minutes to listen to. A good weekly rhythm at four hours: listen on the first day, look over the Visual Companion with a pencil, try the explorer, then spend the remaining time in the workbook.

The level sits between final-year high school and first-year university. Nothing is assumed beyond what you met at school, and it is fine if that was a long time ago. Where a first-year idea appears, it is introduced from the ground up.

In this course

5.1 The sky and the history of astronomy
5.2 Gravity and orbits
5.3 Light as messenger
5.4 The Solar System
5.5 Stars
5.6 Stellar death
5.7 Galaxies
5.8 Cosmology

Astronomy applies the rest of science to the largest scales. Everything we know about stars and galaxies comes from light and gravity, so this course draws heavily on Course 3, especially orbits, waves and modern physics, and on the logarithms and powers of ten from Courses 1 and 2. It is also written from the Southern Hemisphere, where much of the best sky is.

Module 5.1

The sky and the history of astronomy

Long before telescopes, people tracked the Sun, Moon and planets with remarkable precision. Understanding the everyday sky, and how our picture of it changed from Ptolemy to Kepler, is the foundation of astronomy.

By the end of this module you can

- Explain day and night, seasons and the Moon's phases
- Describe the celestial sphere, constellations and Southern Hemisphere sky
- Outline the shift from an Earth-centred to a Sun-centred model

Daily and yearly motions

Earth spins once a day on its axis, so the Sun and stars appear to rise in the east and set in the west. It orbits the Sun once a year, so different constellations are visible at night in different seasons. Astronomers picture the sky as a giant celestial sphere with Earth at the centre: a convenient fiction for describing directions.

Seasons come from the tilt of Earth's axis, twenty-three point four degrees, not from changing distance to the Sun. In fact Earth is closest to the Sun in early January, during the southern summer. When the Southern Hemisphere leans towards the Sun, sunlight strikes it more directly and days are longer, so it is summer there and winter in the north.

The Moon

The Moon always has one half lit by the Sun. Its phases come from how much of that lit half faces us as it orbits Earth every twenty-nine and a half days. Earth's shadow only falls on the Moon during a lunar eclipse. In the Southern Hemisphere, the Moon's phases appear flipped compared with northern pictures: a waxing crescent is lit on the left.

The southern sky

Southern observers see things northerners never do: the Southern Cross, the bright pointers Alpha and Beta Centauri, the Magellanic Clouds, which are small neighbouring galaxies, and the dense central bulge of the Milky Way high overhead in winter. Aboriginal and Torres Strait Islander astronomy, among the oldest continuous traditions in the world, includes the Emu in the Sky, traced in the dark dust lanes of the Milky Way, and uses its position to mark seasons.

From Ptolemy to Kepler

The ancient Greek model, refined by Ptolemy around one hundred and fifty CE, put Earth at the centre with planets on circles riding on circles. It predicted positions reasonably well for over a thousand years. In 1543 Copernicus proposed that Earth and planets orbit the Sun, which explained why planets sometimes appear to move backwards. Tycho Brahe made the most accurate naked-eye

measurements ever, and Johannes Kepler used them to show in 1609 that orbits are not circles but ellipses. Galileo's telescope then revealed moons orbiting Jupiter and the phases of Venus, evidence that not everything circles Earth.

Bridge from the social sciences

The Copernican shift is a classic case study in the history and sociology of science: a better model that initially predicted no better, adopted for reasons of simplicity and later evidence. Kuhn's account of paradigm change drew on it.

Key ideas to carry forward

- Seasons come from axial tilt, not distance
- Phases: how much of the Moon's lit half we see
- Southern sky: Southern Cross, Magellanic Clouds, galactic centre
- Copernicus, Tycho, Kepler and Galileo moved the Sun to the centre

Module 5.2

Gravity and orbits

Newton showed that the force that drops an apple also holds the Moon in orbit. One law of gravity, combined with the laws of motion, explains every orbit in the Solar System and lets us weigh planets and stars.

By the end of this module you can

- State Kepler's three laws
- Use Newton's law of universal gravitation
- Calculate orbital speeds and periods, and the mass of a central body

Kepler's laws

Kepler found three rules. Planets move in ellipses with the Sun at one focus. A line from Sun to planet sweeps out equal areas in equal times, so planets speed up near the Sun. And the square of the orbital period is proportional to the cube of the orbit's size [Equation 5.2-1]. In years and astronomical units, T squared equals a cubed exactly for planets orbiting our Sun.

Universal gravitation

Newton proposed that every mass attracts every other mass with a force proportional to the product of the masses and inversely proportional to the square of the distance between them [Equation 5.2-2]. G is tiny, six point six seven times ten to the minus eleven in SI units, which is why gravity between everyday objects is imperceptible. Henry Cavendish first measured it in 1798, an experiment often described as weighing the Earth.

Near Earth's surface this law gives the familiar g. At the surface, GM over r squared equals nine point eight metres per second squared [Equation 5.2-3]. Go twice as far from Earth's centre and gravity falls to a quarter.

Orbits

An orbit is falling around. Throw a ball sideways fast enough and the ground curves away as fast as the ball falls. For a circular orbit, gravity provides exactly the centripetal force from Module 3.2: G M m over r squared equals m v squared over r. The orbiting mass cancels, so orbital speed depends only on the central mass and the distance [Equation 5.2-4]. This also gives Kepler's third law with the constants filled in [Equation 5.2-5], so measuring a moon's orbit tells you the planet's mass.

Astronauts feel weightless not because there is no gravity in orbit, which is still about ninety per cent of surface strength at the International Space Station, but because they and their craft are falling together. Escape velocity is the launch speed needed to never fall back [Equation 5.2-6]; for Earth it is about eleven kilometres per second.

Key ideas to carry forward

- Kepler: ellipses, equal areas, $T^2 \propto a^3$
- Newton: $F = GMm/r^2$, an inverse-square law
- Orbital speed $v = \sqrt{(GM/r)}$, independent of the orbiting mass
- Measuring an orbit weighs the central body

Module 5.3

Light as messenger

Almost everything we know about the universe beyond the Solar System arrives as light. Its brightness, colour, spectral lines and shift tell us how far away things are, how hot they are, what they are made of, and how they are moving.

By the end of this module you can

- Describe the electromagnetic spectrum and why telescopes are placed where they are
- Use the inverse-square law, Wien's law and spectral lines
- Explain parallax and Doppler shift as tools for distance and motion

The electromagnetic spectrum

Visible light is a narrow slice of the electromagnetic spectrum, which runs from radio waves through microwaves, infrared, visible, ultraviolet and X-rays to gamma rays. All travel at the speed of light and differ only in wavelength. Earth's atmosphere blocks much of the spectrum, which is why X-ray and infrared telescopes go into space, while radio telescopes like those at Parkes and the Square Kilometre Array site in Western Australia work from the ground.

Brightness and distance

Light spreads out over an ever-larger sphere as it travels, so brightness falls with the square of distance [Equation 5.3-1]. A star twice as far away appears a quarter as bright. Astronomers use this, plus the star's true luminosity, to find distance.

For nearby stars, distance comes from parallax: as Earth orbits, a nearby star shifts slightly against the background. The parallax angle is tiny, less than one arcsecond, a three thousand six hundredth of a degree. A star with a parallax of one arcsecond is defined to be one parsec away, about three point two six light-years [Equation 5.3-2].

Colour, temperature and spectra

Hot objects glow, and the hotter they are, the bluer the light. Wien's law says the peak wavelength is inversely proportional to temperature [Equation 5.3-3]. The Sun, at about five thousand eight hundred kelvin, peaks in the visible; red stars are cooler, blue stars hotter.

Passing starlight through a prism reveals dark absorption lines: atoms in the star's outer layers absorbing specific wavelengths, exactly the energy-level fingerprints of Module 3.9. Helium was discovered in the Sun's spectrum in 1868, before it was found on Earth.

Doppler shift

Spectral lines shift towards longer wavelengths if the source is moving away, a redshift, and towards shorter wavelengths if approaching. The shift measures speed along the line of sight [Equation 5.3-4]. It reveals binary stars orbiting each other, planets tugging their stars, and, on the largest scale, the

expansion of the universe.

Key ideas to carry forward

- Telescopes go to space for wavelengths the atmosphere blocks
- Brightness ∝ 1/distance^2; parallax gives distances to nearby stars
- Hotter means bluer: $\lambda_{max} \propto 1/T$
- Spectral lines reveal composition; their shift reveals motion

Module 5.4

The Solar System

Our Solar System formed from a collapsing cloud of gas and dust about four and a half billion years ago. Its structure, rocky worlds inside and giants outside, records how it was made.

By the end of this module you can

- Describe the Sun, the planets and the smaller bodies
- Explain the formation of the Solar System from a disc
- Describe how exoplanets are found

The Sun

The Sun holds more than ninety-nine per cent of the Solar System's mass. It is a ball of hot plasma, mostly hydrogen and helium, powered by nuclear fusion in its core at about fifteen million kelvin. Each second it converts about four million tonnes of mass into energy [Equation 5.4-1]. That energy takes many thousands of years to work its way out, then eight minutes to reach Earth.

The planets

The four inner planets, Mercury, Venus, Earth and Mars, are small and rocky. Venus has a crushing carbon dioxide atmosphere and a runaway greenhouse effect that makes it hotter than Mercury. Mars has ancient riverbeds that show it once had liquid water. Beyond the asteroid belt lie the giants: Jupiter and Saturn, made mostly of hydrogen and helium, and Uranus and Neptune, rich in ices. Distances are measured in astronomical units, the average Earth–Sun distance of a hundred and fifty million kilometres [Equation 5.4-2].

Formation

A rotating cloud of gas and dust collapsed under gravity and flattened into a spinning disc, with the young Sun at its centre. That is why the planets orbit in nearly the same plane and the same direction. Close to the Sun it was too hot for ices, so only rock and metal could clump together, making small rocky planets. Further out, beyond the frost line, ices were plentiful, cores grew larger, and they captured huge amounts of gas, becoming giants. Comets and asteroids are leftover building material, and meteorites let us date the Solar System to four point five six billion years.

Other planetary systems

Since 1995, astronomers have found more than five thousand planets around other stars. Most are detected indirectly: a planet passing in front of its star dims the light slightly, the transit method [Equation 5.4-3], or its gravity makes the star wobble, revealing itself through a Doppler shift. Some systems look nothing like ours, with giant planets orbiting closer than Mercury.

Key ideas to carry forward

- The Sun is a fusion-powered ball of plasma with 99% of the mass
- Inner planets rocky, outer planets giants: the frost line explains it
- The Solar System formed from a spinning disc 4.6 billion years ago
- Exoplanets: transits and Doppler wobbles

Module 5.5

Stars

Stars are balls of gas held together by gravity and held up by the energy of nuclear fusion. A single diagram, plotting brightness against temperature, organises almost everything we know about them.

By the end of this module you can

- Explain the balance between gravity and pressure in a star
- Describe luminosity, the Stefan–Boltzmann law and the HR diagram
- Explain how a star's mass decides its lifetime

A balancing act

A star is a contest between gravity, pulling everything inwards, and pressure from hot gas and radiation pushing outwards. In a stable star they balance exactly. The energy comes from fusion in the core: four hydrogen nuclei combine into one helium nucleus, losing about zero point seven per cent of their mass as energy. Stars form when clouds of gas collapse until the core is hot and dense enough, around ten million kelvin, to start fusion.

Luminosity, temperature and size

Luminosity is the total power a star emits. It depends on surface area and on temperature to the fourth power, the Stefan–Boltzmann law [Equation 5.5-1]. A star twice as hot, with the same size, is sixteen times more luminous. A cool star can still be very luminous if it is enormous: that is a red giant.

The Hertzsprung–Russell diagram

Around 1910, Ejnar Hertzsprung and Henry Norris Russell plotted stars' luminosity against their surface temperature, with hot on the left by tradition. Most stars fall along a diagonal band, the main sequence, where they spend most of their lives fusing hydrogen. Giants and supergiants lie above it, cool but brilliant; white dwarfs lie below, hot but faint. The Sun sits in the middle of the main sequence.

Mass is destiny

A star's position on the main sequence is set by its mass. Massive stars are hotter, brighter and bluer; low-mass stars are cool, faint and red. Massive stars also burn their fuel ferociously: luminosity rises roughly as mass to the power three and a half, so lifetime falls steeply with mass [Equation 5.5-2]. The Sun will last about ten billion years; a star ten times more massive lasts only a few tens of millions; the smallest red dwarfs will outlive the current age of the universe many times over.

Key ideas to carry forward

- Stars balance gravity against pressure from fusion energy
- $L = 4\pi R^2 \sigma T^4$: luminosity depends on size and temperature^4
- HR diagram: main sequence, giants, white dwarfs
- More massive stars are brighter and live much shorter lives

Module 5.6

Stellar death

How a star dies depends on its mass. Sun-like stars swell and shed their outer layers gently; massive stars explode as supernovae, forging the heavy elements in your body and leaving behind neutron stars or black holes.

By the end of this module you can

- Describe the end states of low-mass and high-mass stars
- Explain supernovae and the origin of the elements
- Describe white dwarfs, neutron stars and black holes

Sun-like stars

When a Sun-like star exhausts the hydrogen in its core, the core contracts and heats while the outer layers swell enormously: it becomes a red giant. In about five billion years the Sun will expand past the orbit of Venus. The core eventually fuses helium into carbon and oxygen. Then the outer layers drift off as a glowing shell, misleadingly called a planetary nebula, leaving the bare core as a white dwarf: about the size of Earth but with roughly half the Sun's mass. A teaspoon of it would weigh several tonnes. It slowly cools for billions of years.

Massive stars and supernovae

Stars more than about eight times the Sun's mass go further, fusing successively heavier elements in shells like an onion until the core is iron. Fusing iron does not release energy, so the core can no longer support itself. It collapses in under a second, and the rebound blasts the star apart in a supernova that can briefly outshine a whole galaxy.

Supernovae and colliding neutron stars scatter heavy elements into space, where they are swept into new stars and planets. The calcium in your bones, the iron in your blood and the oxygen you breathe were all made in stars. Carl Sagan's line that we are made of star stuff is literally true.

Neutron stars and black holes

If the collapsed core is between about one point four and three solar masses, it becomes a neutron star: a city-sized ball of neutrons, so dense that a sugar-cube volume weighs about a billion tonnes. Conservation of angular momentum makes them spin rapidly, and some sweep beams of radio waves across Earth like lighthouses: pulsars, discovered by Jocelyn Bell Burnell in 1967.

Heavier cores collapse completely into black holes, regions where gravity is so strong that the escape velocity exceeds the speed of light. The boundary is the event horizon, at the Schwarzschild radius [Equation 5.6-1]. For the Sun's mass it would be only three kilometres. In 2019 the Event Horizon Telescope imaged the shadow of a black hole for the first time, and gravitational waves from merging black holes were first detected in 2015.

Key ideas to carry forward

- Sun-like stars: red giant, planetary nebula, white dwarf
- Massive stars: iron core collapse, supernova
- Heavy elements are made in stars and supernovae
- Remnants: white dwarf, neutron star or black hole, by mass

Module 5.7

Galaxies

Our Sun is one of a few hundred billion stars in the Milky Way, which is one of perhaps two trillion galaxies. Galaxies hold stars, gas, dust and something we cannot see at all.

By the end of this module you can

- Describe the structure of the Milky Way and our place in it
- Classify galaxies by shape
- Explain the evidence for dark matter and supermassive black holes

The Milky Way

The pale band across a dark sky is our own galaxy seen edge-on from inside. It is a barred spiral about a hundred thousand light-years across, with a central bulge, a thin disc with spiral arms where new stars form, and a sparse halo of old stars and globular clusters. The Sun lies about twenty-six thousand light-years from the centre and takes around two hundred and thirty million years to complete one orbit [Equation 5.7-1].

At the very centre sits a supermassive black hole, Sagittarius A star, of about four million solar masses. Its mass was measured by tracking stars orbiting it, using Kepler's third law, work that earned the 2020 Nobel Prize.

Kinds of galaxy

Edwin Hubble sorted galaxies by shape. Spirals have discs and arms, rich in gas and young blue stars. Ellipticals are smooth, rounded collections of old red stars with little gas. Irregulars, like the Magellanic Clouds, have no clear shape. In the 1920s Hubble also settled a great debate by measuring the distance to the Andromeda nebula and showing it was a separate galaxy far outside our own. Andromeda, two and a half million light-years away, is heading towards us and will merge with the Milky Way in a few billion years.

Dark matter

In the 1970s Vera Rubin and Kent Ford measured how fast stars orbit in spiral galaxies. If most mass were in the visible stars, outer stars should orbit more slowly, as outer planets do. Instead the rotation curves stay flat far out. There must be extra unseen mass, dark matter, about five times more than ordinary matter. Gravitational lensing and the motions of galaxies in clusters tell the same story. What dark matter is remains one of the great open questions.

Key ideas to carry forward

- The Milky Way is a barred spiral; the Sun is 26 000 ly from the centre
- Galaxy types: spiral, elliptical, irregular
- Supermassive black holes sit at galactic centres
- Flat rotation curves reveal dark matter

Module 5.8

Cosmology

Cosmology asks the biggest questions: how did the universe begin, how has it changed, and what will become of it? Remarkably, observations now answer much of this with numbers.

By the end of this module you can

- Explain Hubble's law and the expansion of the universe
- Describe the evidence for the Big Bang
- Outline dark energy and the universe's composition and fate

An expanding universe

In 1929 Edwin Hubble, building on measurements by Vesto Slipher and Henrietta Leavitt, found that distant galaxies are receding, and the further away, the faster: velocity is proportional to distance [Equation 5.8-1]. The constant of proportionality, the Hubble constant, is about seventy kilometres per second per megaparsec.

This does not mean we are at the centre. Picture dots on a balloon being inflated: every dot sees every other dot moving away, faster for those further away. Space itself is stretching, and the light travelling through it is stretched to longer wavelengths: the cosmological redshift.

The Big Bang

Run the expansion backwards and everything was once closer, denser and hotter. Dividing distance by speed gives a rough age [Equation 5.8-2], and careful measurements give thirteen point eight billion years. Three pillars of evidence support this hot Big Bang. First, the expansion itself. Second, the cosmic microwave background, a faint glow filling the sky, discovered by accident in 1965: light released when the universe cooled enough for atoms to form, about three hundred and eighty thousand years after the start, now stretched into microwaves at two point seven kelvin. Third, the abundance of light elements: about three quarters hydrogen and one quarter helium by mass, exactly what nuclear physics predicts for the first few minutes.

What the universe is made of

In 1998 two teams measuring distant supernovae found that the expansion is accelerating. Something is pushing space apart: dark energy. Combining all observations, ordinary matter, everything in stars, planets and people, makes up only about five per cent of the universe's energy content. Dark matter is about twenty-seven per cent, and dark energy about sixty-eight [Equation 5.8-3].

If dark energy behaves as it seems to, the universe will expand forever, growing colder and darker as stars burn out over trillions of years. But both dark matter and dark energy are names for things we do not yet understand. Cosmology remains one of the most exciting frontiers in science.

Bridge from the social sciences

Cosmology is a historical science: like archaeology or epidemiology, it cannot rerun its subject, and it relies on converging lines of evidence. The 'three pillars' argument is a strong example of consilience.

Key ideas to carry forward

- Hubble's law: $v = H_0 d$; space itself is expanding
- Age $\approx 1/H_0 \approx 13.8$ billion years
- Big Bang evidence: expansion, microwave background, light-element abundances
- Universe: 5% ordinary matter, 27% dark matter, 68% dark energy